

InvestCircle — How It Works

Tech stack & flow, in plain language, for the team

React+Vite

Express

PostgreSQL

JWT

Vercel

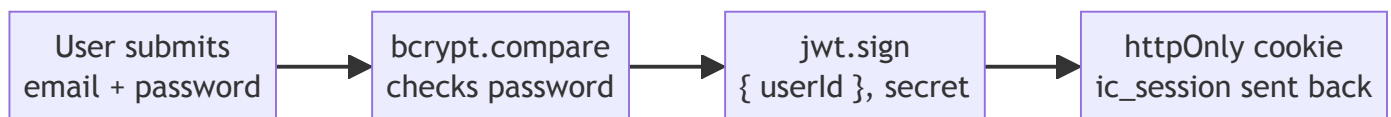
1. The big picture

The browser (frontend) never talks to the database directly — it always asks the Express server, and only the server reads/writes the database. That's why passwords and business rules live on the backend, never in code shipped to the browser.



Layer	Choice	Why
Frontend	React+Vite, Tailwind, React Router	Fast dev server, minimal config
Backend	Node.js + Express	One small server, easy to read
Database	PostgreSQL, plain SQL (no ORM)	Just SQL, nothing extra to learn
Auth	bcrypt + JWT in a cookie	No session table to manage
Hosting	Vercel, one project, both halves	Same domain → no CORS/cookie issues

2. Logging in — the JWT flow

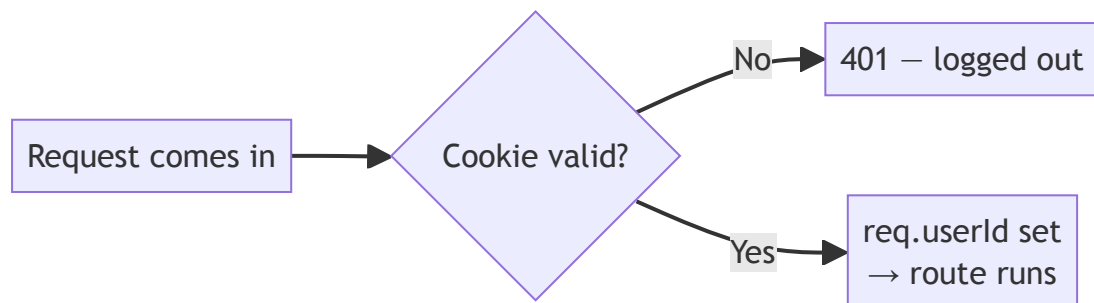


Password: never stored as text — only a one-way `bcrypt` hash. Even a stolen database can't be reversed back into passwords.

JWT: a signed token holding just the user's id. Signed = server can prove it made it; nobody can forge one without the secret key.

Why not a session table? Verifying a JWT is pure math (check the signature) — no database lookup needed on every request. One less table to keep in sync.

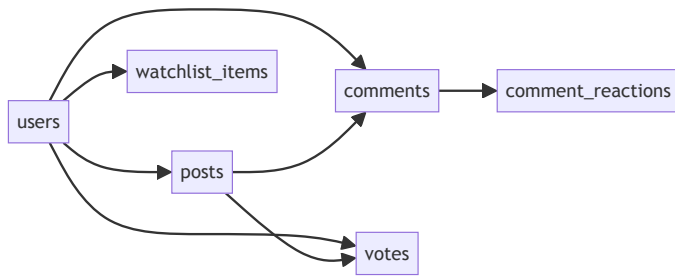
3. Every request after login



The browser attaches the cookie automatically to every request (frontend just sets `credentials: 'include'`). No token handling in frontend code at all.

`requireAuth` blocks the route if the cookie's missing/invalid. `optionalAuth` never blocks — used for read-only pages. **Logout** just deletes the cookie — nothing to clean up server-side.

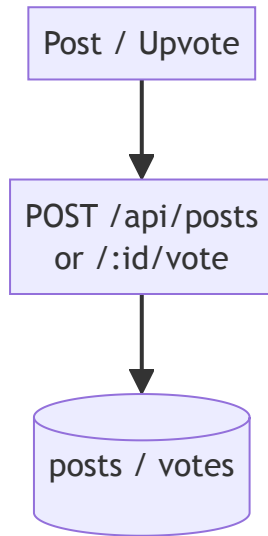
4. The database



One PostgreSQL database (backend/schema.sql), tables linked by IDs. **No ORM** — hand-written SQL with \$1, \$2 placeholders (blocks SQL injection). **No migration tool** — CREATE TABLE IF NOT EXISTS runs on every startup. Counts (votes, reactions) are computed live with SUM() , never stored as a running number.

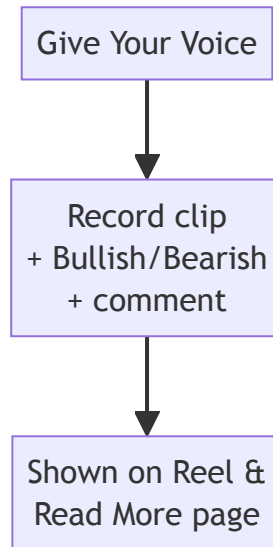
5. How each feature works

Discussions



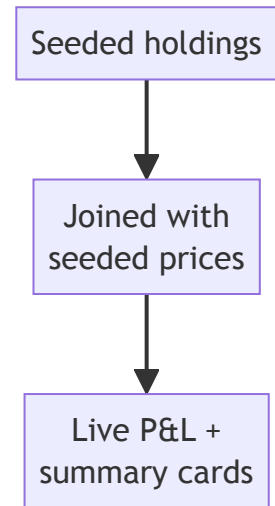
Voting is an upsert — one vote row per user per post.

News Reel



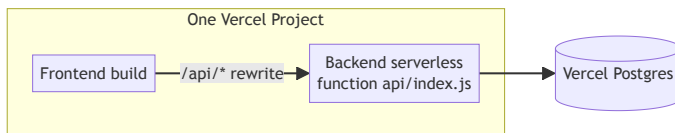
Recording stays in the browser — kept simple for the demo.

Portfolio



No live market feed by design — clearly labeled demo data.

6. Deployment & codebase layout



Frontend — organized by feature, not file type:

- features/ — auth/, discussions/, news/, portfolio/
- shared/ — components/, context/, lib/ (used by 2+ features)

Backend mirrors it by resource: routes/auth.js, posts.js, news.js, watchlist.js — one file per feature.

Same domain for both halves → the login flow above works identically in dev and production; only DATABASE_URL and JWT_SECRET differ.

InvestCircle — Software Engineering group project